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Water-management trade-offs in Lomellina rice paddies

**Evidence from Castello d'Agogna and the pilot irrigation district
in San Giorgio di Lomellina**

Literature-based critical technical report

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Executive summary

This report assesses the effect on the system (ecosystem services) of three common irrigation strategies (mainly continuous flooding, delayed flooding and water-seeded alternate wetting and drying). These were being applied in a linked paddy-canal-groundwater system. This report keeps its focus on local evidence and data. The major sources of data and evidence are field trials in Castello d'Agogna, hydrological modelling for the San Giorgio di Lomellina pilot district and groundwater monitoring in the Province of Pavia. The map reveals evidence anchors (municipality level). It does not represent a legal limit of Lomellina, or the exact boundaries of the field experiment or irrigation district.

Under the conditions tested, water-seeded alternate wetting and drying (AWD) resulted in clear combined climate impact and irrigation savings during the growing season, when being compared with the continuous flooding. This was evident as reported source for $\text{CH}_4\text{--N}_2\text{O}$ GWP₁₀₀ was 46% lower for AWDsafe and 54% lower for AWDstrong than WFL. Mean net irrigation declined from 1,351 mm to 1,006 mm (−25.5%) and 932 mm (−31.0%), respectively. Pooled grain yields were around 10.3, 10.4 and 10.6 Mg ha^{−1}. The pooled means mask cultivar-specific responses, differences between the gas and yield measurement sets, and the grain-quality trade-off. AWD reduced As concentrations but increased Cd and Ni (Vitali et al., 2024).

The scale dependence of management impacts is summarized here. Modelled water supply from April to September for the San Giorgio pilot district is lower for wet-seeded AWD and delayed flooding (DFL) than WFL. Modelled supply of 19.7 million m³ decreased to 16.5 million m³ and 16.1 million m³ under wet-seeded AWD and DFL, respectively. In June, demand rose by 24.5% under DFL. The district AWD scenario applies a −10 kPa trigger at 5 cm, which is not representative of either AWDsafe (−5 kPa at 5 cm) or AWDstrong (−20 kPa at 5 cm). Similarly, lowering net irrigation in the fields does not demonstrate the equal lowering of consumptive use when it flows back into the canal-groundwater system as in the form of seepage and drainage.

Most likely progression is to trial AWDsafe in monitored local area and not convert the whole district. Trial needs treatment-level replication and monitor annual $\text{CH}_4\text{--N}_2\text{O}$ balance, cultivar-specific yield, grain As/Cd/Ni, field inflows and outflows, groundwater persistence and selected aquatic indicators. Future district planning based on the external evidence showing that alternative irrigation regimes may not uniformly improve but can redistribute services, should trial spatial mosaics based on beneficiary and service goals.

1. Background, aims and boundaries

Agriculture practices have been developed and evolved for many years in the rice paddies which are commonly based and linked over characteristics of floodplain cropping system network. Major characteristics are limited oxygen diffusion, changes in microbial metabolism, temporary aquatic phase, the interaction and links among the field storage, canals and shallow ground water. Greenhouse gas fluxes, nutrient dynamics, recharge and return flow, grain production, food quality and the habitat potential may all be affected by the changing hydroperiod. Which functions change, where and when, and for whom is an important issue for the realistic assessment of system services.

The key question is: What are the conditions under which different flooding practices lead to lower methane emissions and irrigation needs in the studied rice system of Lomellina, while ensuring good food production performance, and how do these conclusions differ at field scale and irrigation district scale?

This section organizes the discussion around three questions:

Q1. What do local experiments tell about greenhouse gas emissions, irrigation need, yield and grain quality? Q2. How does the role of groundwater exchanges, return flows and seasonal delivery times impact the interpretation of water savings at field scale?

Q3. What conclusions remain sustainable regarding experimental design, annual accounting boundaries and geographical transferability?

Spatially bound results are from the two locations in the Province of Pavia, with data: Castello d'Agogna experimental platform and ca. 1,000 ha pilot irrigation district (western part of the municipality of San Giorgio di Lomellina). Data for field comparison locally were for 2012-2013 and 2021-2022 and were different studies as it was not a direct time comparison. Groundwater monitoring covers 2017-2018 and 2018-2019 winters and 2018 summer. Satellite context cover approximately 2000-2016. Field data are local effects estimates, which are not being provided by the external studies. But they do test annual and northern Italian data.

This report is a focused critical review with secondary analysis and original mapping. It does not estimate and reproduce the extent of irrigation practices, repeat the San Giorgio Pilot district model, deduce a formal Lomellina boundary, or put monetary values on anything on already existed data and findings. It assumes that yield of grain implies food supply and is a constraint on possible management fluctuations. Indicators, beneficiaries, and accounting boundaries are needed for water volumes, recharge, greenhouse-gas fluxes, metal(loid)s, and hydroperiod before these can be considered services or disservices.

2. Study system and management alternatives

2.1. Ecological characteristics and spatial context

The focus of the study is not just one paddy field but rather a interconnected paddy-canal-groundwater network as demonstrated by the evidence and data collected at field and district levels. It interacts with the surrounding paddy fields during the growing season via rice fields, unlined canals, drainage structure and with shallow aquifer. Water entering a field could end up being used by nearby field or another user

due to infiltration. The scale of this network system within the San Giorgio pilot district covers about 1,000 ha western part of the municipality (Gilardi et al., 2023).

The system boundary used in this review is the managed paddy field -canal – and groundwater network represented by the retained studies. The municipalities' polygons in Figure 1 demonstrate the location of the interested evidence, they do not delimit the experimental plots, the pilot irrigation district or Lomellina as a whole.

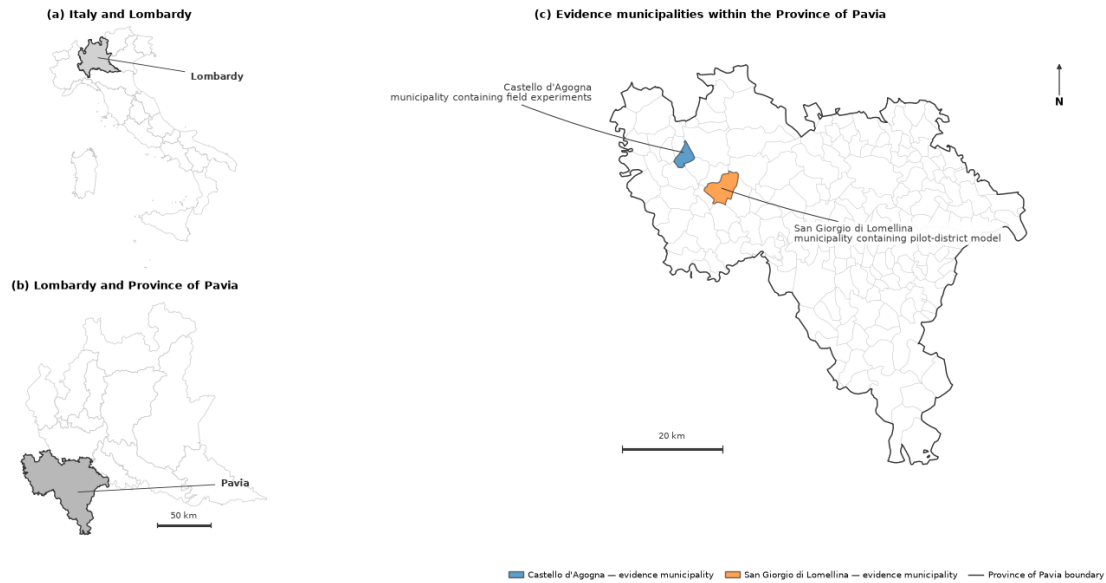


Figure 1. Administrative boundaries: ISTAT, redistributed through Guglielmo/geojson-italy; 1 January 2026 vintage; WGS84/EPSSG:4326. Castello d'Agogna and San Giorgio di Lomellina are the municipality-level anchors. Neither is a polygon of an experimental plot, neither represents the pilot district or any official border of Lomellina. Map prepared for this report.

Loam topsoil (pH 5.6; organic carbon 11.3 g kg⁻¹ in the upper 30 cm) is the context for the 2021-22 field experiment at Castello d'Agogna (Vitali et al., 2024). Groundwater position, organic substrate, acidity and texture decide the soil drying, redox recovery and solute mobility. They usually represent specific sites and not the regional averages. Results reported for different treatments should be understood as referring to a specific interconnection of soil-crop-water.

The hydroperiod usually provides a typical seasonal ecology cycle. Early flooding creates shallow water habitat and connects fields with conveyance channels. Mid season drainage or AWD interrupts the aquatic cycle and starts the oxic period. Finally, harvest drainage ends the aquatic cycle. The drying schedule due to the crop calendar avoids damaging crops. A critical soil water level is achieved depending on combined canal management, rainfall, evaporation, soil storage and groundwater head and how long it lasts. The same pathway applies to biological processes. Rice is the managed primary producer. Carbon and nitrogen cycles are controlled by microbial guilds including methanogens, methanotrophs, nitrifiers and denitrifiers responding to the available oxygen. Wetlands system of paddies and canals provide temporary habitats for aquatic insects and other organisms in paddies and canals that can move back and

forth. (Bo et al., 2025). Their activity depends on hydroperiod duration, refuge opportunities and water quality, that cannot be predicted only on the basis of irrigation volumes.

On a wider scale, Ranghetti et al. (2018) shows a change in irrigation timings. They analyzed northwestern Italian rice surfaces based on MODIS data for the period 2000-2016 and found evidence of changes in spring flooding. These data set the scene but can't talk about the treatment thresholds, fields water balances or greenhouse gas responses for the two focus locations.

2.2. Management alternatives require operational definitions

Irrigation labels sometimes can conceal nuances. WFL might include regular drainage; AWD varies among crops, drying threshold and depth, and protected crop stage. DFL shifts when floods begin and therefore the peak needs of the season. Table 1 consider these aspects as part of the management alternative rather than a footnote.

Table 1. Operational definitions of the retained irrigation families.

Management family	Establishment and water regime	Operational detail	Interpretive limit
CF/WFL	Water establishment followed by predominantly flooded cultivation; operational drains may occur.	Reference schedule must be taken from the source.	Not equivalent to uninterrupted inundation.
DFL	Flooding begins later; one local gas study used dry seeding and intermittent irrigation with a -30 kPa threshold at 10 cm.	Establishment date changes early recharge and crop exposure.	Field and district DFL protocols are related, not identical.
AWDsafe	Water-seeded; irrigation resumes at -5 kPa measured at 5 cm.	Tested at Castello d'Agogna in 2021–2022.	Field result only; no matching district simulation.
AWDstrong	Water-seeded; irrigation resumes at -20 kPa measured at 5 cm.	More severe drying than AWDsafe.	Field result only; cultivar and grain-quality constraints apply.
Wet-seeded (AWD model)	Wet establishment; intermittent irrigation after tillering at -10 kPa measured at 5 cm.	San Giorgio pilot district for 2013–2020.	Cannot be attributed specifically to AWDsafe or AWDstrong.

Sources: Vitali et al. (2024), Peyron et al. (2016), and Gilardi et al. (2023). CF = continuous flooding; WFL = water-seeded flooding; DFL = delayed flooding; AWD = alternate wetting and drying.

3. Ecosystem functions and service logic

3.1. Redox regulation, methane and nitrogen

Floodwater significantly reduces the transfer of oxygen into the soil. By limiting the oxygen diffusion into the soil it promotes the reducing conditions for oxygen and as result favors methanogenesis, which release the methane. The release of methane depends on the production, oxidation and transport through the soil, water, and soil and rice plants. During the drying season, methane production declines because of the more oxygen transfer into the soil called as reoxygenating. Its production also depends on the substrate availability, temperature, crop stage and length of aeration (Peyron et al., 2016; Vitali et al., 2024).

However, the dynamics of Nitrogen differ somewhat. Nitrification favors aerations while denitrification happens with reflooding and anaerobic microsites. Which might produce or consume N_2O . Research studies in the field of rice in Italy find the complicated interactions between water status, fertilizer timing and carbon availability for nitrogen transformation in the early season (Verhoeven et al., 2019).

Low CH_4 production does not necessarily suggest or linked with low N_2O or integrated climate impact. That is the reason each climate comparison respect the boundaries of source accounts. During the growing season CH_4 - N_2O GWP₁₀₀, annual field balance and life-cycle assessment all have the different process representations. Different number also depend on the other conversion factors.

This report does not aggregate effects by mixing studies using incomparable timeframes or GWP factors. It just compares sign and size within one study.

3.2. Water balance, reuse and temporal reliability

Field balance $\Delta S = R + Q_{in} - Q_{out} - ET - G$, where ΔS is the change in field storage, R is rainfall, Q_{in} and Q_{out} are surface and lateral inputs and outputs, ET is evapotranspiration, and positive G represents net downward groundwater exchange. These terms and equation follow the field-scale accounting described by Facchi et al. (2018).

This is conceptual accounting identity not a fitted equation and it provides no additional estimates. Reduction in the field input can be the impact from either reduced percolation or drainage and consequent reductions in return flow, canal recharge or downstream availability of water. Timings and total amounts may matter with the monthly limits imposed by district operations unit. Only recharge that is available where and when required will be valuable for example a temporary increase in the ground water level after winter may be insufficient for the summer crop production. Where the target species duration, connectivity and conservation goals need to be taken into account to make a credible service claim

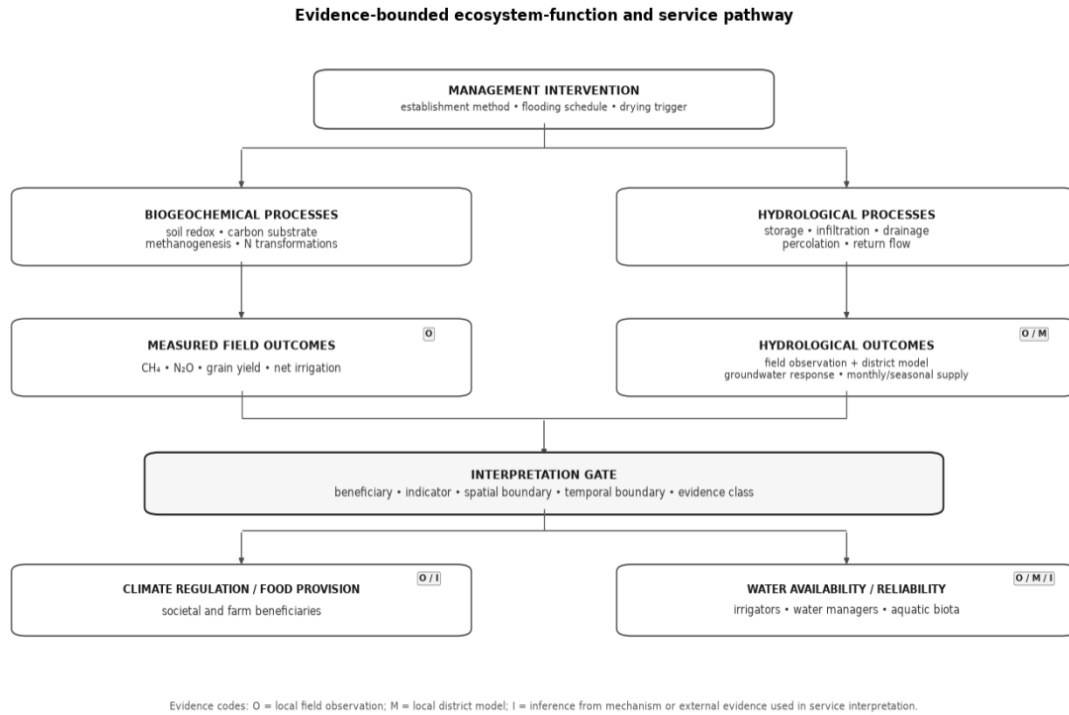


Figure 2. Function-to-service pathway. Irrigation management changes biogeochemical and hydrological functions; measured or modelled outcomes acquire service or disservice meaning only after the beneficiary, indicator, spatial boundary, temporal boundary, and evidence class are specified. Original synthesis based conceptually on Fisher et al. (2009) and the retained rice studies.

3.3. Indicators, beneficiaries and evidence classes

Table 2 distinguishes between measured/modelled results and how those results are interpreted as a service.

Observed outcomes from the focal field experiments or hydrology monitoring. Modelled is the result of a model run of San Giorgio pilot district scenario. Inferred is a mechanism-to-service interpretation or a source located outside the geographical scope of the study. Confidence declares inference, not the impact factor of the journal.

Table 2. Outcome-to-service interpretation and evidential confidence.

Measured/modelled outcome	Indicator	Service interpretation and beneficiary	Evidence class	Confidence
Lower growing-season climate burden under water-seeded AWD	CH ₄ -N ₂ O GWP ₁₀₀ ; % versus WFL	Climate regulation for society and climate-policy beneficiaries	Observed locally	Moderate

Measured/modelled outcome	Indicator	Service interpretation and beneficiary	Evidence class	Confidence
Lower field net irrigation	mm	Potential delivery or scarcity benefit; not demonstrated as consumptive saving	Observed locally; service inferred	Moderate for outcome; low for service
Changed district supply and monthly peak	million m ³ , Apr–Sep and June	Potential reliability and operating benefit/cost to irrigation organisations	Modelled locally; service inferred	Moderate for scenario
Groundwater-level response after flooding	response magnitude and persistence	Potential seasonal water availability to users or dependent ecosystems	Monitored locally; service inferred	Low–moderate
Maintained pooled grain yield	Mg ha ⁻¹ at 14% moisture	Food provision; farm income requires price and cost data	Observed locally	Moderate
Changed grain As, Cd and Ni	grain concentration	Food-quality trade-off affecting consumers, farmers and the value chain	Observed locally	Moderate
Changed hydroperiod	inundation duration and continuity	Potential habitat benefit or disservice for aquatic taxa and conservation actors	Mechanistic/local inference plus external evidence	Low

Sources: Vitali et al. (2024) for growing-season GWP₁₀₀, net irrigation, pooled grain yield, and grain As, Cd, and Ni concentrations; Gilardi et al. (2023) for district water-supply scenarios; Negri et al. (2020) for groundwater response and persistence; Bo et al. (2025) for aquatic-habitat context; and Fisher et al. (2009) for the distinction between ecosystem functions, services, and beneficiaries. Service interpretations and confidence ratings are the author's synthesis. Observed = locally measured evidence; modelled = local model output; inferred = mechanism-based or externally supported interpretation.

4. Review and analytical methods

4.1. Search, eligibility and extraction

This review was completed on 4th of September 2026 and was selectively confirmed in sources on 6th of September. Exact title, DOI and topic combinations were being used as search queries on Google scholar, peer reviewed journal publisher sites (Elsevier ScienceDirect, Wiley Online Library, MDPI and frontiers), University of Milan AIR and IRTA repositories. Backward reference search was used to confirm design and over all mechanistic background. Typical searches, sources and limitations were

being cited in Appendix A. Due to lack of full hit counts, duplicate screening, protocol registration and dual reviewer selection. It is a traceable targeted review rather than a systematic review. Local rice irrigation, topic related hydrology, greenhouse gas mechanisms, yield or grain quality limits, system service multifunctionality or temperate comparison crucial to test boundary conditions are topic covered by included papers. No strict data filtering was applied to the date of publication. Articles for 2009 to 2026 are retained. The record retained includes 17 peer-reviewed papers and one journal correction. Geography area, year of studies, method of establishment, threshold for drying, winter types, independent treatment replications, target population, units, temporal limits and locations of each crucial and relevant statement are among the information being retained and used. Preference was usually given to full text. Data and Evidence presented at abstract level was used explicitly for contextual interpretation and did not transfer external effect size to Lomellina.

4.2. Critical appraisal and numerical treatment

Appraisal considers the directness of focal system, treatment level replication/model calibration, outcome compatibility, stated uncertainty and study consistency. 'Moderate' suggests directly pertinent local evidence with the material limitations in the replication, measurement population or validation; 'low-moderate' is partially local or indirect evidence; 'low' implies mechanism based or out of area assumption but no direct local measurement. These labels were used for information only and are not used to derive an overall quality assessment. Percentage changes were calculated in this report as $100 \times (\text{alternative} - \text{reference}) / \text{reference}$, using the rounded source means cited in Appendix B. Values were taken at the published mean, annual values, units and signs. The annual markers in Figure 3 show the 2021 and 2022 source values; they are not confidence intervals or independent treatment-level replicates.

Uncertainty values were not reconstructed. No new or additional statistical or meta analysis were performed. District estimates are subject to the model. Discharge and groundwater performance were calibrated using 2015-2016 observations and simulated scenarios of management in 2013-2020, reported in Gilardi et al. (2023).

The Calibration fit is not independent validation of out of sample results. The longer simulation period does not constitute locally adopted performance. Corrected percolation values are based on Mayer et al. (2020), not the parent paper. Numerical data is ambiguous, excluded: gas estimates are assumed to be inventory calculations and not chamber observations in Gharsallah et al. (2023).

5. Results

5.1. Q1—field climate, irrigation and production results

The most suitable study for the water seeded AWD is the 2021–2022 Castello d'Agogna comparison (Figure 3). The mean emission factor of methane over two years were 2.54, 1.27 and 1.08 kg CH₄ ha⁻¹ d⁻¹ for WFL, AWDsafe and AWDstrong. The mean net irrigation of these fields was 1,351, 1,006 and 932 mm, which corresponded to –25.5% and –31.0% relative changes with the two AWD treatments. The pooled grain yields were 10.3, 10.4 and 10.6 Mg ha⁻¹ (Vitali et al., 2024).

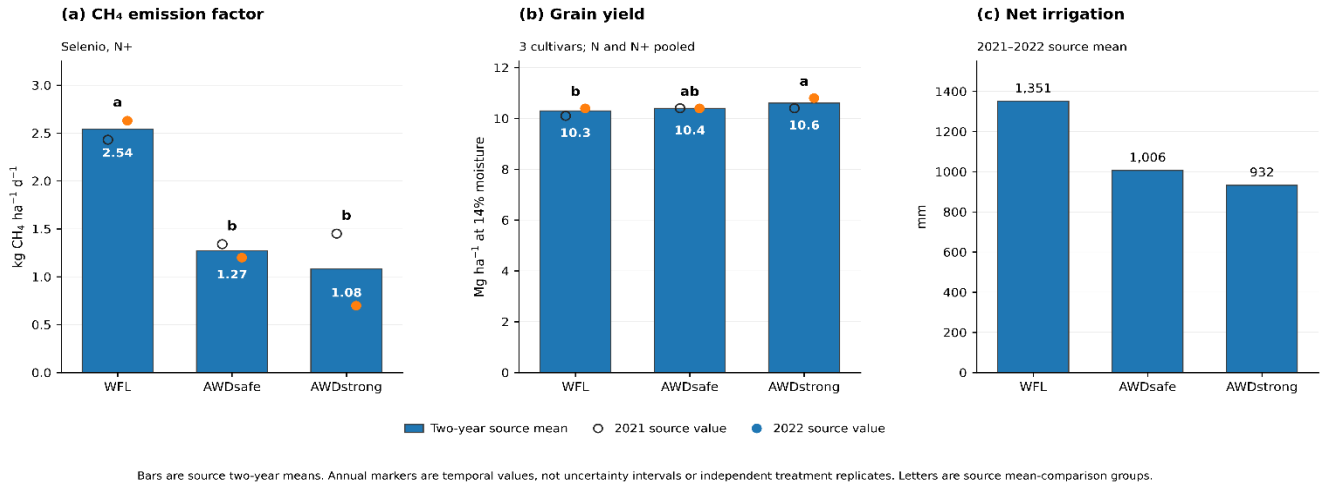


Figure 3. Local field information from Vitali et al. (2024), Tables 2 and 6 and Section 2.2. Bars are two-year means from the source; markers are 2021 and 2022 data. Gas is Selenio at N+; yield is across three cultivars and N/N+. The letters showing the comparison between groups are taken from the source and are not uncertainty ranges.

Stable yield should be interpreted carefully. The source comparison letters indicate that AWDstrong differed significantly from WFL, whereas AWDsafe belonged to the overlapping ab group and therefore did not differ significantly from either treatment.

The interaction between variety and water management was significant. There were two main plots for each irrigation treatment system. Gas sampling used four chambers, with two nested within each plot. Selenio was the crop grown with N+ fertilization. The outcomes of the yield demonstrate the means of three cultivars and two N and N+ treatments. It is also worth remembering that the plots have different management, but the same crops. The two chambers nested within each of the two main plots per treatment are subsamples. They do not create four independent treatment-level replicates.

Grain composition is another production quality factor. Total As decreased, but Cd and Ni significantly increased with AWD relative to WFL, increasing with AWD severity for Cd and recorded only in 2022 for Ni (Vitali et al., 2024). These findings are reported in the paper. However, unreported uncertainty or regional exposure has not been inferred here.

A second methane response can be found in local DFL data. Peyron et al. (2016) record 59% less methane and 56% less annual GWP₁₀₀ compared to WFL, but only one plot per water treatment had gas chambers. Treatments and fields are confounded. This outcome is consistent with the aeration mechanism. But it does not establish a region-wide causal effect.

Another study by Carrijo et al. (2017) collates 56 studies and 528 comparisons: mild AWD reduced total water use by 23.4% and severe AWD reduced yield by 22.6%. Such heterogeneous global estimates are boundary evidence, not local predictions for the focused region in this study.

5.2. Q2—district demand and groundwater timing

The seasonal amount in the San Giorgio pilot district deviate from the peak demand (Figure 4). In comparison to the WFL, wet seeded AWD and DFL decreased modeled April to September supply from 19.7 to 16.5 and 16.1 million m³ (−16.2% and −18.3%). June figures were 4.9 to 3.5 million m³ (−28.6%) for AWD and 6.1 million m³ (+24.5%) for DFL (Gilardi et al., 2023).

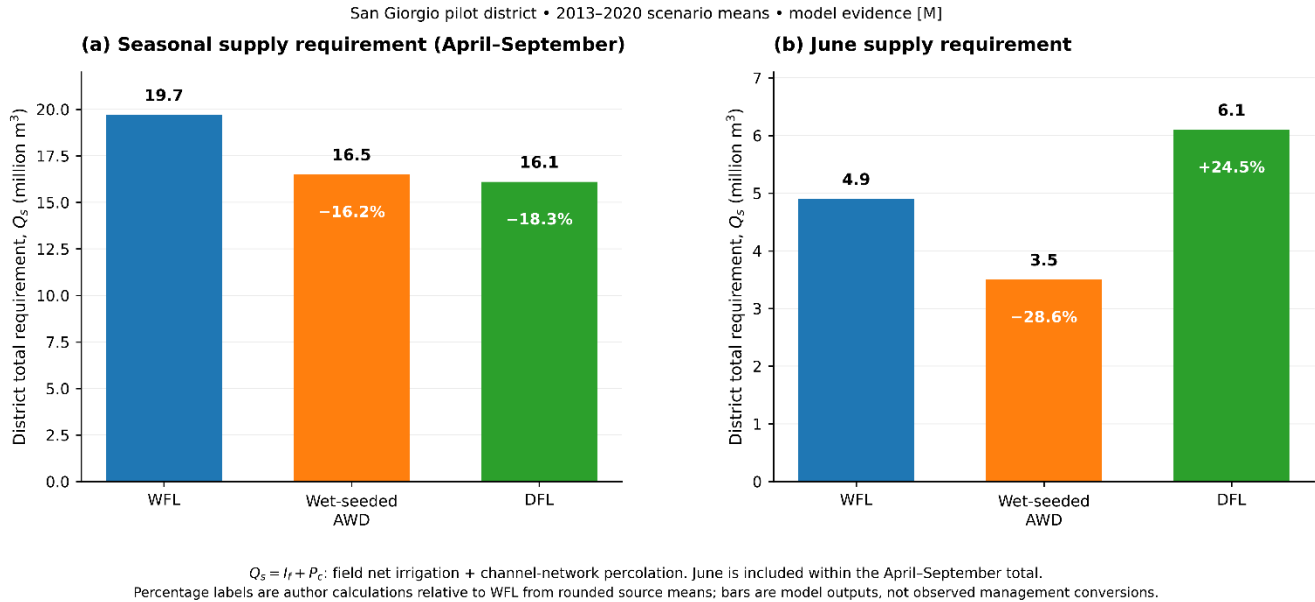


Figure 4. Modelled water-supply requirement for the San Giorgio pilot district, 2013–2020 means, redrawn from Gilardi et al. (2023). June is included within April–September. Percentages were calculated from the rounded source means and are shown to one decimal place.

This difference occurs because of protocol mismatch. The wet seeded AWD with intermittent irrigation following tillering by Gilardi's method with -10 kPa at 5 cm differs from the wet seeded AWD safe and strong with intermittent irrigation starting after tillering by Vitali's methods with -5 kPa and -20 kPa at 5 cm. The related Mayer model also has a dry seeded WDA and a different time frame. All these results share a common set of assumptions and share a history and district background for the district, a set of model and hence are not the independent replications (Mayer et al., 2019, 2020; Gilardi et al., 2023).

Groundwater observations also help to clarify a second time constraint. Usually, groundwater recovers to its pre flood level within about 1 month after floods stops. Increase in groundwater from winter flooding were found by Negri et al. (2020) in the three Pavia areas, but summer flooding takes normally longer time for groundwater recovery. Different responses arise from the different flooding sizes and from the neighboring flooding and soils. The recharge volume and practical duration should be assessed separately. This data show a local hydrologic response, not a general estimate of avoided shortage.

5.3. Q3—annual climate and multifunctionality boundaries

The Climate balance of growing season is being reported for the local water seeded AWD study and the dry fallow extension for peyron's comparison. There are also other external indicators which shows why this distinction matters. Table 3 retains the time horizon and GWP factors of each study, rather than trying to combine incommensurate results.

Table 3. Climate findings and their accounting boundaries

Study and setting	Change in GWP ₁₀₀ relative to reference	Boundary and interpretation
Vitali et al. (2024), Lomellina	AWDsafe -46%; AWDstrong -54%	Growing-season CH ₄ + N ₂ O; factors 28 and 265.
Peyron et al. (2016), Lomellina	DFL -56%; dry intermittent irrigation -83%	Annual field measurements with dry winters; factors 25 and 298; one gas-instrumented plot per treatment.
LaHue et al. (2016), California	AWD treatments -57% to -74%	Annual field-gas assessment including flooded fallow; factors 34 and 298; external system.
Echeverría-Progulakis et al. (2025), Ebro Delta	AWD +26.8% in combined annual estimate. The combined annual AWD effect was not statistically significant ($\chi^2 = 1.9$, $p = 0.393$).	Growing-season field measurements plus flooded-fallow mesocosms; factors 27 and 273; external and not field-only annual evidence.

LaHue et al. (2016) used three randomized blocks in California, measured by flooded fallow periods. The annual GWP₁₀₀ remained 57-74 % less than the reference over AWD treatments. In the Ebro Delta, Echeverría-Progulakis et al. (2025) used a combination of growing season field trials, a post harvest flooded fallow mesocosm and estimated a 26.8 % annual increase in response to AWD. However, the treatment effect was not statistically significant ($\chi^2 = 1.9$, $p = 0.393$). Neither study predicts the Lomellina annual balance, but both demonstrate that a growing season outcome does not provide certainty on an annual claim.

This Phenomenon continues in a two year Ebro Delta field experiment, taking it to multifunctionality. It compare the continuous flooding, mid season drainage and AWD and finds similar overall multifunctional performance but differentiated service provision. CH₄ was around 80 kg C-CH₄ ha⁻¹ less under AWD and mid season drainage. However, this advantage for climate regulation came with a decrease in soil carbon storage of 8.8%. Grain yield was less, vertebrate abundance less, and some invertebrate diversity increase with AWD (Echeverría-Progulakis et al., 2026). This is not carried forward to Lomellina. It implies the type of effect size where an apparently favorable outcome on one service turn to disadvantage when others are included into consideration.

6. Integrated interpretation and management implications

6.1. Evidence-coded trade-off synthesis

Irrigation schemes redistribute functions rather than moving all outcomes only in one direction. As in table 4 retains the distinction between local observations (O), local model outputs (M) and mechanisms or external evidence (I). There are no weights, money totals or a general management scores.

Table 4. Practice-by-outcome synthesis relative to the predominantly flooded reference.

Outcome	CF/ WFL	DFL	Water-seeded evidence AWD
Climate	Higher local CH ₄ reference (O)	Lower local annual GWP ₁₀₀ ; weak treatment replication (O)	Lower local growing-season GWP ₁₀₀ ; annual balance unresolved (O)
Field irrigation	Reference input	Not directly compared in Figure 3	Lower net irrigation at Castello d'Agogna (O)
District supply	Reference seasonal and June demand (M)	Lower seasonal demand; higher June peak (M)	Lower seasonal and June demand under related -10 kPa scenario (M)
Hydrological regulation	Earlier inundation supports recharge and reuse (I)	Delayed initial recharge and supply timing (I/M)	Wet establishment retained; later drying alters exchange timing (I/M)
Food provision	Reference yield and composition (O)	Companion-study context only	Maintained pooled yield, cultivar interaction; As ↓, Cd/Ni ↑ (O)
Aquatic habitat	Longer inundation opportunity (I)	Early-season hydroperiod contraction (I)	Local response unmeasured; external responses are taxon-specific (I)
Primary uncertainty	Higher field irrigation input and methane burden; consumptive-water effect unresolved	Treatment replication and June reliability	Annual GWP ₁₀₀ , metal(loid)s, groundwater persistence and habitat

O = local field observation; M = local district model; I = inference from mechanism or external evidence. Sources: Vitali et al. (2024), Peyron et al. (2016), Miniotti et al. (2016), Gilardi et al. (2023), Facchi et al. (2018), Negri et al. (2020), and Echeverría-Progulakis et al. (2026). The district AWD model uses -10 kPa at 5 cm and is not AWDsafe at -5 kPa.

There are three management options. DFL lowers the modeled seasonal district total but it does delay early inundation and peaks modeled demand in June. CF/WFL has the highest number in methane

emissions and water supply requirements in the retained comparison in April-September. whereas DFL has the highest June requirement. Earlier and longer hydroperiod might be compatible with recharge timing, return flow and aquatic habitat opportunity. Water seeded AWD offers the strongest local combination of less growing season climate damage, less field irrigation and high pooled yield. Grain composition was measured local level. However, food safety implication across cultivars, soil conditions and repeated years remain unresolved. One of the precautionary candidates for more local testing is AWDsafe at -5 kPa and 5 cm. Because its -5 kPa threshold is less severe while retaining substantial growing season GWP and irrigation reductions. This is a management judgement, not a statistically demonstrated optimum. It is not yet a district recommendation. One signal district can't optimize climate regulation, water reliability, food quality and biodiversity due to heterogenous nature of soil and groundwater conditions. The idea of patchwork approach, keeping some flooded or delayed flooding zones when recharge timings or habitat is explicit and testing AWD where the benefits for fields are repeatable, is a research proposal, not a proved best practice.

6.2. Robustness and limitations

Strongly restricting the data does not change the key results and findings. The removal of abstract Miniotti et al. (2016) record removes the companion DFL – yield context; the removal of the California and Ebro studies removes the external annual contrast and multifunctionality warning in findings. These data restrictions does not regulate the local methane, field irrigation, district demand or groundwater timings results. There is a special case of monitored water seeded AWD trail (for example pilot) in the focal field sites. Several restricts and checks prevents the stronger claims on the local causality or regional applicability. There is limited independent replication of local gas treatments. There are non-identical gas, yield and grain quality populations. There are modeled district scenarios based on a non matching AWD trigger. Groundwater monitoring which does not measure a district scale reliability benefits. There are no annual AWD emissions and local aquatic response measurements. Quantitative analysis without compatible and accurate raw data would mislead about the precision of existing summaries

6.3. Monitoring and decision framework

Logically , the next step would be to replicate CF/WFL, DFL and AWDsafe in another field experiment, including comparing soil and groundwater environments. Treatment assignment should define experimental units, not chambers or quadrats. Measure need to be all on the same treatment population: gas, yield, grain chemistry and hydrology over the relevant fallow period, monitoring inflows, outflows, evapotranspiration, groundwater levels, nitrogen pathways, trigger adherence and selected aquatic measures. Decision metrics should be driven by beneficiaries. For farmers, it is essential to know yield variation by cultivar, labor, energy, water costs and grains As/Cd/Ni. Volume per season, monthly peak and losses due to conveyance reuse downstream (irrigation manager); location and duration of recharge (water manager); response of taxa in paddies, canals and refuges (conservation). In all other cases, a composite metric will mask as much as it reveals about trade-offs.

6.4. Valuation and implementation

Valuation should follow the establishment of the biophysical chain. The value of delivered water may be assessed from marginal costs of delivery, pumping, treatment or shortage, distinguishing between

lowering depletion and return flow. For climate regulation, the time specific carbon damage value could be used for the consistent range of CO₂- equivalent changes that keep the individual components CH₄, N₂O and soil carbon intact.

For farm value, treatment specific gross margins should be computed as the yield $\times$ farm- gate price less water, energy, labour, monitoring and transition costs.. The quality of the grain harvest should be assessed separately from the yield. Initially, habitat and recharge should remain in biophysical terms linked to particular beneficiaries. Monetary benefit transfer would be defensible only where ecological comparability and the affected beneficiary population can be demonstrated. There is a local example of multi domain assessment by Gharsallah et al. (2023). While Fisher et al. (2009) supports distinction of the different functions of the ecosystem and eventually its benefits. Replicated field trials need to be followed by a district pilot operational phase before any stakeholders weighting or monetary comparison.

7. Conclusions

Q1. Water-seeded AWD at Castello d'Agogna reduced source reported growing season GWP₁₀₀ by 46% for AWDsafe and 54% for AWDstrong. Net irrigation fell by 25.5% and 31.0%. Yields were 10.4 and 10.6 Mg ha⁻¹ versus 10.3 Mg ha⁻¹ for WFL. These pooled results do not remove cultivar interactions or differences among the gas, yield and grain composition measurement population: As decreased while Cd and Ni increased.

Q2. Lower field irrigation input does not guarantee an equivalent hydrological-service benefit. In the San Giorgio pilot district seasonal demand and June peak demand did not change in the same direction. DFL lowered the seasonal total but increased June demand by 24.5%. The wet seeded AWD scenario reduced both seasonal and June supply requirements. But its -10 kPa trigger cannot be attributed to either AWDsafe at -5 kPa or AWDstrong at -20 kPa. Groundwater monitoring likewise distinguishes a recharge response from its persistence.

Q3. This supports a monitored AWDsafe pilot based on local conditions but not immediate district-wide conversion or a universal ranking of irrigation regimes. Uncertainty on annual balance of CH₄-N₂O, groundwater persistence for the exact field protocol, grain quality over soil types and cultivar, and aquatic habitat bears on the decision process. Professional recommendations are: staged, explicitly monitored experimentation. Then district analysis testing service bundles against named beneficiaries and seasonal limits.

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Appendix A. Review traceability and evidence appraisal

The searches were iterative and purposive. Table A1 is a record of representative searches conducted during the report writing process from literature review to findings. After identifying the most relevant candidates exact titles and DOI were checked.

Table A1. Representative executed searches and selection boundaries.

Query	Purpose	Selection boundary
"Lomellina" rice methane irrigation groundwater alternate wetting drying	Locate integrated local evidence.	Retain local studies; distinguish measurements from models.
"rice" "delayed flooding" "yield" "Italy"	Assess DFL production evidence.	Use only accessible findings; do not invent missing statistics.
"rice" "winter flooding" "groundwater" "Italy"	Assess recharge persistence.	Separate winter monitoring from growing-season trials.
"rice" "alternate wetting and drying" "fallow" emissions	Test annual climate offsets.	Identify field versus mesocosm designs and winter regime.
"Rice yields and water use" Carrijo 2017	Locate yield/water synthesis.	Use as global context, not a local prediction.
"rice" multifunctionality "alternate wetting and drying" ecosystem services 2026	Identify multi-service evidence.	No quantitative transfer from the Ebro Delta.

Table A2. Principal evidence records and retained uses.

Record	Design and boundary	Critical appraisal and retained use
Vitali 2024; full text	Castello d'Agogna, 2021–2022; two main plots per irrigation treatment; gas chambers nested two per plot.	Moderate local evidence. Gas: Selenio, N+; yield: broader population. Grain As/Cd/Ni retained. No annual AWD claim.
Peyron 2016; full text	Same centre, 2012–2013; one gas-instrumented plot per water treatment; dry winters.	Treatment and plot identity not separable. Annual DFL direction retained with qualification.
Miniotti 2016; abstract	Companion agronomic publication for 2012–2013 platform.	Qualitative DFL yield context only; not independent of Peyron.

Record	Design and boundary	Critical appraisal and retained use
Gilardi 2023; full text	San Giorgio pilot district; scenarios 2013–2020; calibration 2015–2016; wet-seeded AWD at –10 kPa, 5 cm.	Moderate for scenario mechanism; no observed district conversion or independent out-of-sample validation.
Mayer 2019 + 2020 correction; full text	Related district model, 2013–2016; dry-seeded WDA.	Lineage and protocol context. Corrected percolation values supersede the parent table.
Facchi 2018; full text	Four adjacent paddies and field-to-field exchanges.	Used for boundary interpretation; no numerical extraction.
Negri 2020; full text	Three Pavia areas, 1.2/36/85 ha; winters 2017–2018 and 2018–2019; summer 2018.	Moderate for site-specific groundwater response; not a regional service estimate.
Carrijo 2017; full text	56 studies/528 comparisons.	External benchmark; heterogeneous protocols and irrigation-plus-rainfall metric.
LaHue 2016; full text	California; three randomised blocks; 2012–2015, including flooded fallow.	External annual field contrast, not a local effect size.
Echeverría-Progulakis 2025; full text	Ebro Delta; growing-season field measurements plus postharvest mesocosms.	Annual boundary evidence; combined design limits transferability.
Echeverría-Progulakis 2026; full text	Ebro Delta, 2022–2023; 19 indicators across biodiversity and ecosystem services.	External trade-off structure; no Lomellina effect-size transfer.
Gharsallah 2023; full text	Castello d’Agogna, 2019–2020; hydrology, modelling and multi-domain indicators.	Context only; ambiguous numerical reporting excluded; gas estimates not treated as measurements.
Ranghetti 2018; full text	MODIS flooded-area assessment, northwestern Italy, 2000–2016.	Regional inundation context; no local treatment outcome.

Additional framework records: Fisher et al. (2009), Verhoeven et al. (2019), Bo et al. (2025), and Zhao et al. (2024). Guglielmo (2026) supplies the ISTAT-derived cartographic data only and is not treated as ecological evidence. Confidence refers to the report’s inference, not overall article or publications journal quality.

Appendix B. Graph data and calculations

Tables B1 and B2 reproduce the numerical inputs used in Figures 3 and 4. The values are published summaries rather than raw observations. Reported pooled means are retained rather than recomputed from rounded annual entries.

Table B1. Figure 3 data from Vitali et al. (2024), Tables 2 and 6 and Section 2.2.

Indicator and period	WFL	AWDsafe	AWDstrong
Methane EF, 2021	2.43	1.34	1.45
Methane EF, 2022	2.63	1.20	0.70
Methane EF, two-year mean	2.54 a	1.27 b	1.08 b
Grain yield, 2021	10.1	10.4	10.4
Grain yield, 2022	10.4	10.4	10.8
Grain yield, two-year mean	10.3 b	10.4 ab	10.6 a
Net irrigation, two-year mean (mm)	1,351	1,006	932

Table B2. Figure 4 data from Gilardi et al. (2023), Figure 9; 2013–2020 scenario means.

Indicator	WFL	Wet-seeded model	AWD	DFL
April–September requirement (million m ³)	19.7	16.5		16.1
June requirement (million m ³)	4.9	3.5		6.1
Seasonal change (%)	0	–16.2		–18.3
June change (%)	0	–28.6		+24.5

The wet-seeded AWD model uses –10 kPa at 5 cm. Changes use $100 \times (\text{alternative} - \text{WFL}) / \text{WFL}$. For example, DFL June change = $100 \times (6.1 - 4.9) / 4.9 = +24.49\%$, reported as +24.5%. Small differences may reflect rounded source means.